# Russel Osiemo

**Data Scientist | Machine Learning, Forecasting and Applied AI**

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## Professional Summary

Data scientist with 4 years of hands-on experience in machine learning, statistical modelling and analytics, including nearly 3 years building production data and ML solutions for a Kenyan FMCG distributor. Takes solutions from problem definition through validation to scheduled production runs, working closely with sales, finance and operations teams. Strong in Python, R and SQL, with practical experience in feature engineering, model evaluation, data quality monitoring and documentation. Has worked with bank transaction, payment, receivables and customer data, including customer segmentation, lifetime value and churn tracking.

## Core Skills

- **Machine Learning and Statistics:** supervised learning (regression, classification), gradient boosting (XGBoost), time-series forecasting (Prophet), hypothesis testing, feature engineering, model evaluation and benchmarking
- **Generative AI and NLP:** LLM intent classification with rule-based guardrails, quantised small language models, spaCy, sentiment analysis, fuzzy entity matching (RapidFuzz)
- **Programming and Data:** Python (pandas, scikit-learn, TensorFlow, PySpark), R, SQL (PostgreSQL, MySQL), MongoDB, Apache Spark
- **MLOps and Deployment:** GitHub Actions CI/CD, scheduled production pipelines, Docker, FastAPI, Flask, Apache Airflow, REST API integration, Git
- **Model Governance:** validation benchmarks, data quality and anomaly monitoring, human review steps, reproducible code, model and pipeline documentation, data privacy controls
- **Visualisation and Communication:** Power BI, Tableau, Streamlit, automated Excel and PowerPoint reporting for management

## Work Experience

**Kingdom Trading Limited and Kingdom Sourcing Limited, Kenya | Data Analyst | Feb 2024 - Present**

- Built Prophet and XGBoost demand forecasting models on full sales history and put them into weekly use for reorder planning, reducing stockout periods by 25%.
- Took automated order processing from problem definition to production: purchase orders are read from email, validated, de-duplicated on order and customer keys, priced from a barcode master net of VAT and posted as sales orders to Zoho Inventory on a scheduled GitHub Actions pipeline.
- Building a payment remittance pipeline that parses PDF remittances, classifies each payment line into five scenarios and posts draft entries to Zoho Books for finance review before approval.
- Set up data quality monitoring and anomaly alerts across all pipelines, reducing error rates by 35%, and built a three-way stock reconciliation app (system vs rebuilt vs physical count) in Flask.
- Designed a weekly SKU movement classification (Fast, Regular, Slow, SLOB) and a competitor price pipeline that uses fuzzy matching to link about 453 SKUs to retailer listings.
- Cascaded annual revenue targets to rep, customer and SKU level using prior-year actuals as weights, and applied NLP sentiment analysis to customer social media data.

**Liquor End Palace, Nairobi | Data Analyst (Consultant) | May 2025 - Present**

- Built a customer analytics framework with RFM scoring, customer lifetime value and ABC/XYZ classification, and track new, reactivated, returning and churned customers to guide retention.
- Split revenue growth into price and volume effects, showing management that growth was price-driven while volumes were falling.
- Built a supplier price comparison tool that uses a language model to match products across 13 supplier price lists.
- Automate monthly management reporting in Python for the directors, sales and finance teams.
- Designing an API integration that syncs stock, products, customers and invoices from the ERP to the field sales app every hour during peak trading.

**DataLyfe | Volunteer Data Scientist (Remote) | Sep 2022 - Present**

- Architect data ingestion workflows for multi-source healthcare data, with patient privacy controls built into each pipeline stage.
- Improved query performance by 45% on large datasets through partitioning and indexing, and automated preprocessing that cut report generation time by 30%.
- Present model outputs and insights to non-technical stakeholders through Tableau dashboards and plain-language reports.

**Ramana Tech School, Nairobi | Big Data Analyst Intern | Jun 2023 - Sep 2023**

- Built a customer 360 view of bank customers from banking transaction data, processing the data with MongoDB and Apache Spark and presenting it in a Streamlit dashboard.
- Standardised data formats across 5+ disparate systems and designed schemas for both analytical and transactional workloads.

## Selected Projects

**Kenya Clinical Reasoning Challenge | PATH/Zindi | Apr 2025 - Present**

- Built a quantised language model (under 1 billion parameters, under 2GB RAM, under 100ms inference) for clinical reasoning on 400+ clinical vignettes.
- Designed an evaluation framework that benchmarks model outputs against expert clinician responses.

**Natural Language Analytics Assistant (Ask Zoho) | 2026**

- Designing an internal assistant where an LLM only interprets the user's question, while predefined, auditable business logic performs every calculation, so no figure is ever generated by the model.
- Delivering in small milestones, each tested against known results before the next is built.

**Optimised Inventory Reordering System | Kingdom Trading | Jul 2024 - Present**

- Applied time-series forecasting to set dynamic reorder points, cutting excess stock by 20%, with supplier lead times and container constraints built into the model.

## Education

**Maseno University, Kisumu | BSc Computer Science | Dec 2023**

Second Class Honours (Upper Division) | GPA 3.21/4.00 | Coursework: Statistics, Machine Learning, Database Systems, Data Structures

**MODCOM Institute of Technology, Nairobi | Certificate in Software Engineering | May 2019**

## Leadership and Volunteer

- **Board Member, Ultimate Wings Kenya (NGO) | 2018 - Present:** contribute to governance and board decisions.
- **Computer Training Volunteer, Ultimate Wing Rescue Centre, Malindi | Sep 2023 - Dec 2023:** trained 50+ students in Microsoft Office and data literacy.